

Course Description: Academic Year 2026-27

Course Title and Code: Calculus and Ordinary Differential Equations (AS1109)	
Teaching Scheme and no. of hours per week	L-T-P: 3-1-0 (4 hours/week)
Credits	4
Programme(s) and Batch	B.Tech. (All branches), 2026-30
Semester (as per curriculum structure)	I Sem
NCrF Level (4.5/5.0/5.5/6.0)	4.5
Pre-requisite course (Course Code and Title)	None
Desired prior knowledge/skills	Higher Secondary Mathematics
Weightage	Theory 100%
The occupational path it may lead to, if any	Foundation for Engineering, Data Science, AI, Scientific Computing, Analytics.
Course Objective: This course introduces the various forms of calculus useful in diverse engineering areas. The course builds upon the concepts learned in single-variable calculus and extends to the functions of several variables. It also introduces the fundamental concepts of ordinary differential equations (ODEs) and analytic methods for solving ODEs.	
Course Outcomes: On successful completion of this course, the students should be able to: AS1109.1: define and visualize functions of several variables, and analyze them using the concepts of limits, continuity, and differentiability. AS1109.2: apply multivariable differential calculus techniques to solve a range of physical problems including optimization. AS1109.3: evaluate multiple integrals and apply them to calculate areas, volumes, in different coordinate systems. AS1109.4: use the concept of the gradient and its relationship to directional derivatives, divergence, and curl to analyze vector fields and scalar fields geometrically. AS1109.5: formulate first-order ordinary differential equations for applications and solve first-order and higher order ordinary differential equations.	

Syllabus

Differential Calculus - Functions of several variables, limits and continuity, partial derivatives, Euler's theorem (The Mixed Derivative theorem), differentiability and total derivative, linearization, the Chain rule, extreme values, saddle points and its applications.

Integral Calculus - Arc length, quadric surfaces, solids of revolution: surface area and volume, double integrals in Cartesian and polar coordinates, change of order of integration, change of variables (Cartesian to polar and vice-versa). Triple Integrals in Cartesian coordinates.

Vector Calculus - Vector Differentiation: Vector functions and derivatives, directional derivative and gradient vectors, tangent plane, divergence, and curl of a vector field.

Ordinary Differential Equations (ODEs) - Introduction to differential equations, First order exact ODEs, integrating factor. Linear ODEs. Second and higher order linear ODEs with constant coefficients, method of variation of parameters.

Text/Reference Books

1. George B. Thomas and Ross L. Finney, Calculus and Analytic Geometry, 9th Edition, Pearson Education Asia, 2006.
2. Erwin Kreyszig, Advanced Engineering Mathematics, 9th Edition, John Wiley & Sons, 2005.
3. C. Ray Wylie and Louis Barrett, Advanced Engineering Mathematics, 6th Edition, McGraw Hill, 2003.
4. Monty J. Strauss, Gerald L. Bradley, and Karl J. Smith, Calculus, 3rd Edition, Pearson Education India, 2002.

Online Resources

1. NPTEL Course: Multivariable Calculus, IIT Roorkee
<https://nptel.ac.in/courses/111107108>
2. MIT OpenCourseWare: Multivariable Calculus and Differential Equations
<https://ocw.mit.edu>

Evaluation Scheme

Note:

1. *Please write the weightage of the components decided by the Institute concerned. Total weightage shall not exceed total.*
2. *Weightage in any single component shall not exceed 40%. The minimum weightage for end-term examinations (Theory) shall be 30%.*
3. *In case the marks are different from weightage, both are required to be mentioned.*

Following are the evaluation schemes for each Institute as per Gradebook

Institute of Engineering and Technology and Institute of Management

Sr. No	Specifications	Weightage	Marks
1	Class Participation	Nil	Nil
2	Assignment	Nil	Nil
3	Quiz	20	20
4	Mid-Term Examination (Theory)	30	30
5	End-Term Examination (Theory)	30	30
6	Report	Nil	Nil
7	Project (PBL)-assign	20	20
8	Lab Evaluation	Nil	Nil
9	Course Portfolio	Nil	Nil
10	Presentation	Nil	Nil
11	Viva	Nil	Nil
	Total	100	100

Attendance Policy

Every student is expected to put in 100% attendance. However, the student with attendance below 75% in a course, shall be penalized as follows.

- 70.0% to less than 75.0% attendance: A warning letter will be issued.
- 50.0% to less than 70.0% attendance: One grade drop. However, a D grade will remain D.
- Less than 50% attendance: Two grades drop. However, a D grade will remain D.

Note:

1. *This Relaxation in attendance includes participation in all kinds of activities, medical emergencies, participation in placement opportunities, late registrations, disciplinary actions, or any other reasons not mentioned here.*
2. *Students will be responsible for maintaining attendance in the classes and monitoring their attendance online.*

Course Articulation Matrix: (Mapping of COs with POs)

Course Outcomes	PO 1	PO 2a	PO 2b	PO 2c	PO 3a	PO 3b	PO 3c	PO 4a	PO 4b	PO 4c	PO 5a	PO 5b	PO 6	PO 7a	PO 7b	PSO 1	PSO 2
AS1109.1	1				1		1										
AS1109.2	1				1		1										
AS1109.3	1				2	1	1										
AS1109.4	2				2	1	1										
AS1109.5	1				1	2		1									

1- Low Correlation; 2- Moderate Correlation; 3- Substantial Correlation

LESSON PLANNING (Theory)

Lecture	Unit/ Chapter/ Topic	Hours
	Unit 1: Differential Calculus	13
	Review of single-variable functions; Curve sketching	
1	Motivation from Engineering, Introduction to functions of several variables, Domain and range	
2-3	Graphs and level curves, visualization of surfaces; open and closed regions, bounded and unbounded regions	
4-5	Limits of functions of several variables; path dependence	
6	Continuity; properties of continuous functions; engineering examples	
7	Definition of partial derivatives; geometric interpretation	
7-8	Higher-order partial derivatives; applications in engineering and economics	
8-10	Differentiability; total differential; linear approximation; error estimation	
11	Multivariable chain rule; implicit differentiation; applications involving composite functions	
12	Equality of mixed partial derivatives (the Mixed Derivative theorem); examples and counterexamples	
12-13	Local maxima, minima and saddle points; second derivative test; optimization problems from engineering	
	Unit 2: Integral Calculus	10
14	Arc length of plane curves; applications	
15	Quadric surfaces: identification and sketching (order one can decide)	
16	Solids of revolution: surface area; applications	
17	Solids of revolution: volume; applications	
18	Double integrals over rectangular regions; interpretation as area and volume	
19	Double integrals over general regions	
20	Double integrals in polar coordinates	
21	Change of order of integration	
22-23	Change of variables: Cartesian to polar conversion and vice-versa; applications and problem solving	
	Unit 3: Vector Calculus	6
24-25	Vector and scalar functions and Fields. Derivatives	
26-27	Gradient of a scalar field, Rate of change in arbitrary directions; directional derivatives, Geometric meaning of gradient; relationship between gradient and directional derivative	
28	Tangent plane to surfaces, Gradient as surface normal vector, engineering applications	
29	Divergence of a vector field, Curl of a vector field; physical interpretation; solenoidal and irrotational fields	
	Unit 4: Ordinary Differential Equations	11

Lecture	Unit/ Chapter/ Topic	Hours
30	Introduction to differential equations: order, degree, solutions of an ODE, formulate first-order ordinary differential equations for applications	
31-32	First-order exact ODEs: condition for exactness and method of solution	
33-34	Equations reducible to exact form; Integrating factors: standard rules and applications	
35	Linear first-order ODEs	
36	Applications of first-order ODEs (growth/decay, cooling, circuits)	
37	Second and higher order linear ODEs with constant coefficients: homogeneous equations	
38	Cases of the characteristic (auxiliary) equation: real, repeated and complex roots	
39	Non-homogeneous linear ODEs: general solution, complementary function and particular integral	
40	Method of variation of parameters, and course wrap-up	